

Solution Requirements

Date	25 June 2026
Team ID	XXXXXX
Project Name	OptiCrop – AI-Based Crop Recommendation System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Functional Requirements (FR) define what the system should do, while Non-Functional Requirements (NFR) define how the system should perform.

Step 1: Functional Requirements (FR)

S. No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	User Input	Users can enter soil nutrient values (N, P, K), temperature, humidity, pH, and rainfall through the web interface.	High
2	Crop Recommendation	The system predicts and displays the most suitable crop based on the entered parameters.	High
3	External Interfaces	The system connects with the trained Machine Learning model and scaler to generate crop recommendations.	Medium

4	Data Processing	The system validates user inputs, preprocesses the data using the scaler, and processes it through the ML model.	High
5	Result Display	The system instantly displays the recommended crop in a user-friendly format.	High
6	Business Rules	Crop recommendation is based on soil nutrients, temperature, pH, and rainfall using a trained ML model.	Medium
7	Data Privacy	User input is processed securely and is not permanently stored by the application.	High
8	Future Enhancement	The system should support adding new crops, datasets, or Machine Learning models in future versions.	Low

Step 2: Non-Functional Requirements (NFR)

S. NO	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate crop recommendations quickly.	Response time less than 3 seconds.
2	Scalability	The system should support future expansion with additional crops and larger datasets.	Supports future enhancement.
3	Security & Data Privacy	User input should be processed securely without unauthorized access.	Secure processing of user input.
4	Reliability & Availability	The system should provide accurate and consistent recommendations with minimal downtime.	Available 99% of the time.
5	Usability & Accessibility	The application should provide a simple, responsive, and user-friendly interface for farmers.	Easy to use with clear navigation.
6	Maintainability	The system should be easy to update with new Machine Learning models, datasets, and features.	New features can be added with minimal code changes